

March_11 LP

March 11, 2024

```
[1]: from sklearn.datasets import load_iris
      from sklearn.tree import DecisionTreeClassifier
      from sklearn.model_selection import train_test_split
      from sklearn.metrics import accuracy_score
```

```
[2]: #Load the iris dataset
      #Split the dataset into training and testing sets
      #Create a DT classifier
      #Expose the Created DT Classifier into Training dataset
      #Predict the labels of the test dataset
      #Evaluate the performance of the model
```

```
[12]: #Load the iris dataset
      iris = load_iris()
      X = iris.data
      y = iris.target

      #Split the dataset into training and testing sets
      X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25,
      ↪random_state = 42)

      #Create a DT classifier
      model = DecisionTreeClassifier()

      #Expose the Created DT Classifier into Training dataset
      model.fit(X_train, y_train)

      #Predict the labels of the test dataset
      y_pred = model.predict(X_test)

      #Evaluate the performance of the model
      accuracy = accuracy_score(y_test, y_pred)
      print('Accuracy: ', accuracy)
```

Accuracy: 1.0

```
[13]: iris
```

```
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predictive attributes and the class\nAttribute Information:\n    - sepal length
in cm\n    - sepal width in cm\n    - petal length in cm\n    - petal width in
cm\n    - class:\n        - Iris-Setosa\n        - Iris-Versicolour\n
- Iris-Virginica\n\nSummary Statistics:\n\n=====
=====
=====
=====
Min  Max   Mean   SD   Class
Correlation\n=====
=====
=====
=====
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length:  4.3  7.9   5.84  0.83   0.7826\nsepal width:  2.0  4.4   3.05
0.43  -0.4194\npetal length:  1.0  6.9   3.76   1.76   0.9490 (high!)\npetal
width:  0.1  2.5   1.20  0.76   0.9565 (high!)\n=====
=====
=====
=====
\n\nMissing Attribute Values: None\n\nClass
Distribution: 33.3% for each of 3 classes.\n\nCreator: R.A. Fisher\n\nDonor:
Michael Marshall (MARSHALL%PLU@io.arc.nasa.gov)\n\nDate: July, 1988\n\nThe famous
Iris database, first used by Sir R.A. Fisher. The dataset is taken\nfrom
Fisher\'s paper. Note that it\'s the same as in R, but not as in the
UCI\nMachine Learning Repository, which has two wrong data points.\n\n\nThis is
perhaps the best known database to be found in the\npattern recognition
literature. Fisher\'s paper is a classic in the field and\nis referenced
frequently to this day. (See Duda & Hart, for example.) The\ndata set contains
3 classes of 50 instances each, where each class refers to a\ntype of iris
plant. One class is linearly separable from the other 2; the\nlatter are NOT
linearly separable from each other.\n\n|details-
start|\n**References**\n|details-split|\n\n- Fisher, R.A. "The use of multiple
measurements in taxonomic problems"\n Annual Eugenics, 7, Part II, 179-188
(1936); also in "Contributions to\n Mathematical Statistics" (John Wiley, NY,
1950).\n- Duda, R.O., & Hart, P.E. (1973) Pattern Classification and Scene

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Analysis.\n (Q327.D83) John Wiley & Sons. ISBN 0-471-22361-1. See page 218.\n- Dasarathy, B.V. (1980) "Nosing Around the Neighborhood: A New System\n Structure and Classification Rule for Recognition in Partially Exposed\n Environments". IEEE Transactions on Pattern Analysis and Machine\n Intelligence, Vol. PAMI-2, No. 1, 67-71.\n- Gates, G.W. (1972) "The Reduced Nearest Neighbor Rule". IEEE Transactions\n on Information Theory, May 1972, 431-433.\n- See also: 1988 MLC Proceedings, 54-64. Cheeseman et al's AUTOCLASS II\n conceptual clustering system finds 3 classes in the data.\n- Many, many more ... \n\n|details-end|\n',

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